

المزج

$$\mathcal{N}(\mu_1, \sigma_1^2) \cdot \mathcal{N}(\mu_2, \sigma_2^2)$$

$$\propto \exp\left\{-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right\} \cdot \exp\left\{-\frac{(x-\mu_2)^2}{2\sigma_2^2}\right\}$$

$$= \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)x^2 + 2\left(\frac{\mu_1}{2\sigma_1^2} + \frac{\mu_2}{2\sigma_2^2}\right)x + c\right\}$$

match $\uparrow$

$$\propto \exp\left\{-\frac{(x-\mu_3)^2}{2\sigma_3^2}\right\} = \exp\left\{-\frac{x^2}{2\sigma_3^2} + \frac{2\mu_3 x}{2\sigma_3^2} + c'\right\}$$

matching : $\frac{1}{\sigma_3^2} = \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \Rightarrow \sigma_3^2 = \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)^{-1}$

$$\frac{\mu_1}{\sigma_1^2} + \frac{\mu_2}{\sigma_2^2} = \frac{\mu_3}{\sigma_3^2} = \mu_3 \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)$$

$$\mu_3 = \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right)^{-1} \left(\frac{\mu_1}{\sigma_1^2} + \frac{\mu_2}{\sigma_2^2}\right)$$

Extending to d-dim. Gaussian dist.

$$\mathcal{N}(\underline{\mu}, \Sigma) \propto \exp\left\{-\frac{(\underline{x}-\underline{\mu})^T \Sigma^{-1} (\underline{x}-\underline{\mu})}{2}\right\}$$

$$\Sigma_3 = (\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}$$

$$\underline{\mu}_3 = (\Sigma_1^{-1} + \Sigma_2^{-1})^{-1} (\Sigma_1^{-1} \underline{\mu}_1 + \Sigma_2^{-1} \underline{\mu}_2)$$

□

$$\tilde{\mu}_1 = \sqrt{1-\beta_t} x_{t-1}$$

$$\mu_2 = \sqrt{\alpha_{t-1}} x_0$$

$$\tilde{\Sigma}_1 = \beta_t I \rightsquigarrow x_t | x_{t-1} \sim \mathcal{N}(\sqrt{1-\beta_t} x_{t-1}, \beta_t I)$$

$$\Sigma_2 = (1-\bar{\alpha}_{t-1}) I \rightsquigarrow x_{t-1} | x_0 \sim \mathcal{N}(\sqrt{\alpha_{t-1}} x_0, (1-\bar{\alpha}_{t-1}) I)$$

$$\begin{aligned}
 x_t | x_{t-1} &\sim c \exp \left\{ - \frac{\|x_t - \sqrt{\alpha_t} x_{t-1}\|^2}{2\beta_t} \right\} \\
 &= c \exp \left\{ - \alpha_t \left\| x_{t-1} - \frac{1}{\sqrt{\alpha_t}} x_t \right\|^2 / 2\beta_t \right\} \\
 &= c \exp \left\{ - \left\| x_{t-1} - \frac{1}{\sqrt{\alpha_t}} x_t \right\|^2 / 2 \left(\frac{\beta_t}{\alpha_t} \right) \right\}
 \end{aligned}$$

$$\Rightarrow \mu_1 = \frac{1}{\sqrt{\alpha_t}} x_t$$

$$\Sigma_1 = \frac{\beta_t}{\alpha_t} I \rightsquigarrow q(x_t | x_{t-1})$$

$$\mu_2 = \sqrt{\bar{\alpha}_{t-1}} x_0$$

$$\Sigma_2 = (1 - \bar{\alpha}_{t-1}) I \rightsquigarrow q(x_{t-1} | x_0)$$

$$\begin{aligned}
 \Sigma_3 &= \left\{ \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) I \right\}^{-1} = \left\{ \frac{\beta_t + \alpha_t (1 - \bar{\alpha}_{t-1})}{\beta_t (1 - \bar{\alpha}_{t-1})} \right\}^{-1} I \\
 &= \left\{ \frac{1 - \bar{\alpha}_t}{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})} \right\}^{-1} I \\
 &= \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} I
 \end{aligned}$$

$$\mu_3 = \Sigma_3 \left[\Sigma_1^{-1} \mu_1 + \Sigma_2^{-1} \mu_2 \right]$$

$$= \left[\frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \right] \cdot \left[\frac{\alpha_t}{1 - \alpha_t} \cdot \frac{1}{\sqrt{\alpha_t}} x_t + \frac{1}{1 - \bar{\alpha}_{t-1}} \cdot \sqrt{\bar{\alpha}_{t-1}} x_0 \right]$$

$$q(x_{t-1} | x_t, x_0) \propto \mathcal{N}(\mu_1, \Sigma_1) \cdot \mathcal{N}(\mu_2, \Sigma_2)$$

$$\propto \mathcal{N}(\mu_3, \Sigma_3)$$

$$= \mathcal{N} \left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} x_0 \right)$$

$$, \quad \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{(1-\bar{\alpha}_t)} \mathbf{I} \quad)$$

$$D_{KL} (q(x_{t-1} | x_t, x_0) \parallel \underbrace{P_\theta(x_{t-1} | x_t)}_{\mathcal{N}(\mu_\theta(x_t, t), \Sigma_3)})$$

$$= \frac{\|\mu_3 - \mu_\theta\|^2}{2\sigma_3^2} = \left\| \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t} \underline{x_0} \right. \\ \left. - \underbrace{\mu_\theta(x_t, t)}_{\epsilon \bar{x}_t + \epsilon_t(\underline{x_t}, t)} \right\|_2^2 / 2 \left(\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \right)$$